

Health & Safety

Standard Radiation protection for Civil Works activities



Reference

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The radiation protection standards presented in this document serve as a practical guide to define the means and prevention measures to be implemented to cope with radioactivity risk.

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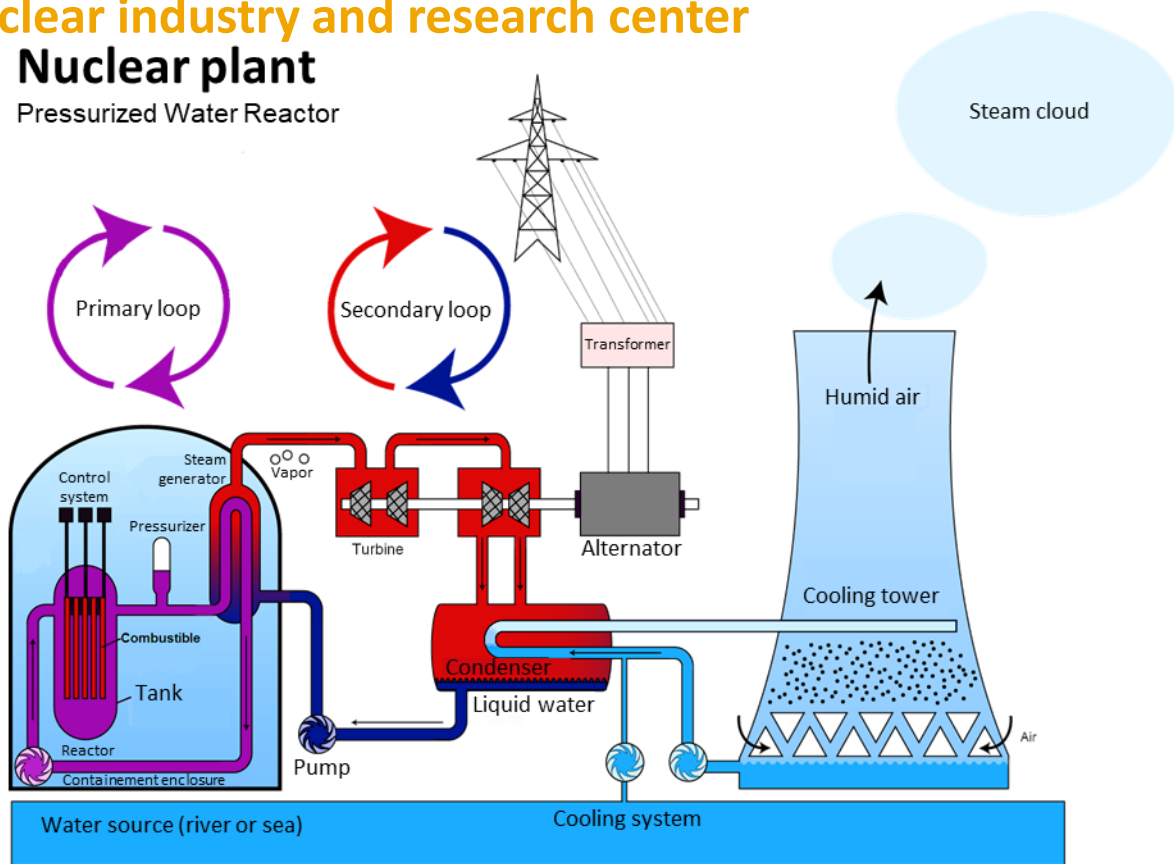
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1. Origin of radioactivity

Nuclear industry and research center

Nuclear plant

Pressurized Water Reactor



Risk to be considered during:

- Maintenance work
- Dismantling, sanitation and waste management work

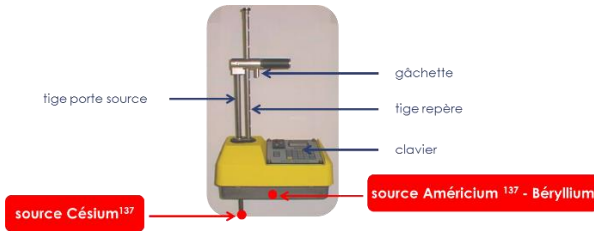


Use of radioactive or X-ray sources

- Density and humidity measurements (e.g. earthworks, slurry from tunnel boring machines)
- R&D earthworks laboratory



Gamma densimeter on mud pressure tunnel boring machine pipes



Gamma densimeter & humidity meter (earthworks)



X-ray laboratory analysis device

2. Radioactivity

Radioactivity is a physical phenomenon of emission of energy in the form of radiation (particle or electromagnetic) by an unstable atomic nucleus (called radioactive).
It is a spontaneous and random phenomenon that can be natural or artificial.

Power of penetration into matter of ionizing radiation

Alpha particles: helium nuclei ($^4\text{He}^{2+}$)

Very low penetration

A sheet of paper is enough to stop helium nuclei

Beta particles: negatively charged electrons

Low penetration (passes through one to two centimeters of living tissue)

Travels a few meters in the air - Aluminum foil a few millimeters thick can stop electrons

Gamma rays and X-rays

Very penetrating type of radiation

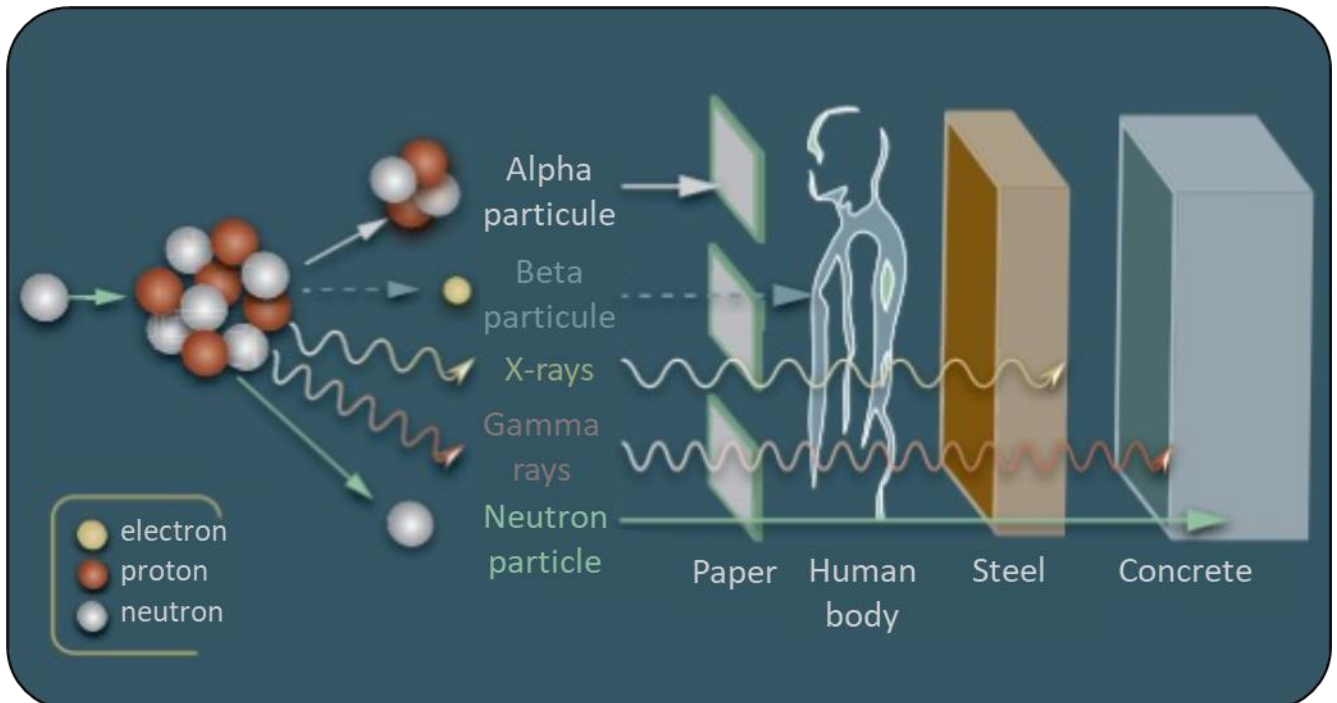
Can scatter underds of meters in the air and passes through anything less dense than a thick block of steel

A thick layer of concrete or lead (or other dense mateiral) provides protection

Neutron particles (requires atomic fission or nuclear fusion)

High penetrating power - Can scatter underds of meters in the air

A high thickness of concrete, water or paraffin stops neutrons

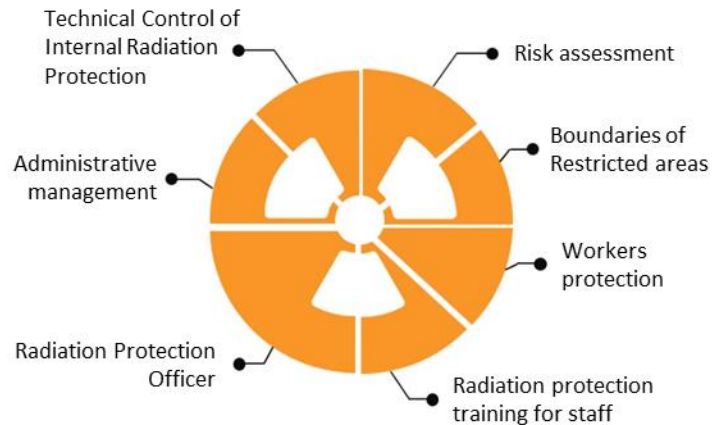


3. Prevention of radioactivity risk

Means of prevention

Radiation protection is all the measures taken to protect humans and their environment from ionizing radiation (IR). Its aim is to ensure that the dose received by people is as low as possible.

At least one competent person in radiation protection is designated as Radiation Protection Officer (RPO) and must have the means necessary to carry out his missions.

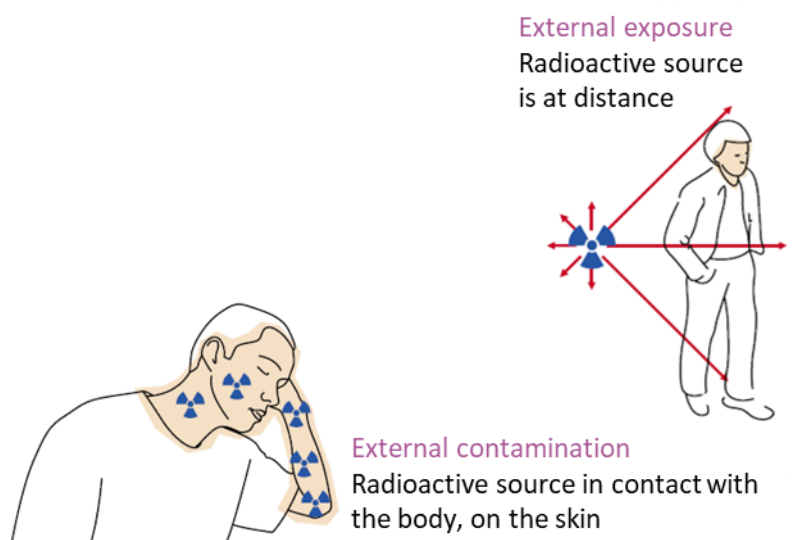
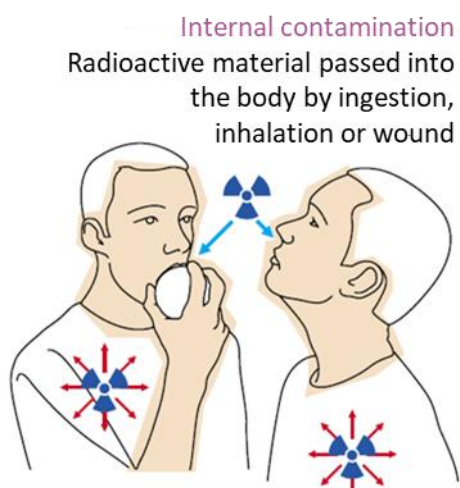


Access to the workstation under ionizing radiation is regulated according to radiological exposure (expressed in Sieverts (Sv)). The limits are set by radiological zoning.

Identify the hazard

The hazard linked to a radioactive source is due to the damage of ionizing radiation to the body's cells. This interaction can occur remotely (irradiation) or directly in contact with the body (external or external contamination).

The dose of ionizing radiation integrated by workers is expressed in Sievert (Sv).



4. Radiological zoning

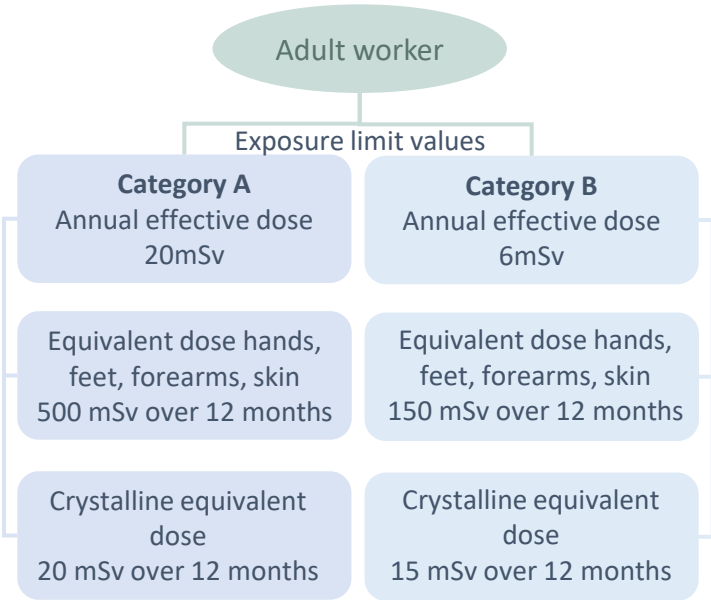
Any more restrictive local regulations must be applied

Dose limits for exposed workers

Workers exposed to ionizing radiation likely to receive an effective dose exceeding 1 mSv over 12 rolling months are classified into 2 categories, A or B:

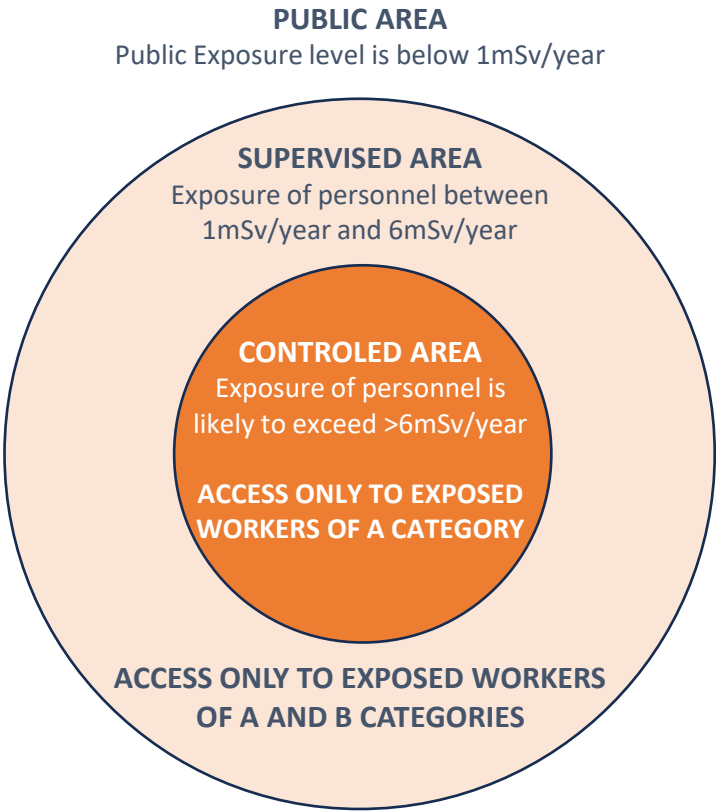
Category	A	B
Equivalent dose (Impact of radiation on body)	> 150mSv	> 50mSv
Effective dose (Impact on the affected tissue/organ)	> 6 mSv	> 1 mSv

- Each exposed worker benefits from:
- appropriate dosimetric monitoring (passive, operational dosimetry)
 - reinforced medical surveillance



Restricted areas

Temporary workers (temporary workers, interns) are prohibited from entering the Controlled area.



5. Access to regulated areas

Required to work in regulated areas

- Medical Aptitude specific to category A or B
- Training in prevention of radiological risk according to the context of the activities (examples: Nuclear clearance risk prevention level 1 or 2, Nuclear Reactor, Fuel Cycle, Research Center)
- Additional training prescribed by the Operator during intervention on a nuclear site
- Wearing individual dosimeters adapted to radiation

Monitored area (from 80 $\mu\text{Sv/month}$ to 1.25 mSv/month)

- Wearing a delayed reading dosimeter is mandatory
- It is possible to use an operational dosimeter (direct reading and alarms)



Controlled area ($> 1.25 \text{ mSv/month}$)

- Wearing the delayed reading dosimeter and the operational dosimeter is mandatory

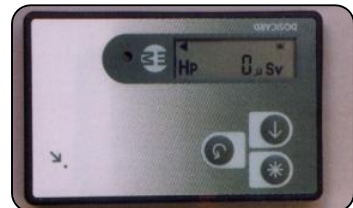


DMC MGP

SAPHYDOSE gi



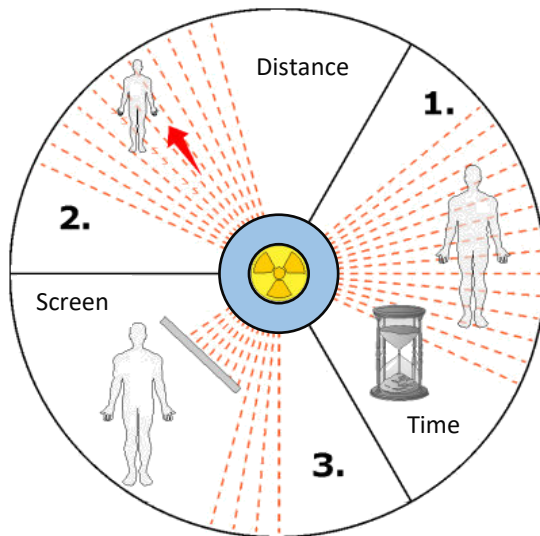
PND-BDT bubble
neutron dosimeter



DOSICARD

6. Means of collective and individual protection

Protection from External Exposure

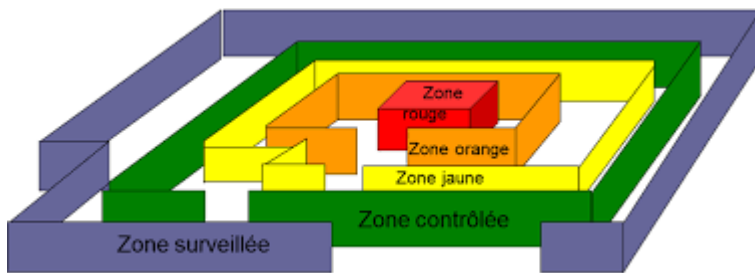


Screens



Remote work

Protection from External Contamination



- Surface control and cleaning
- Personal Protection: overalls, gloves, etc.

Internal Exposure Protection



Containment of sources, sealed circuits, etc.



Respiratory protection, waterproof clothing, etc.

7. Means of control and monitoring – 1/2

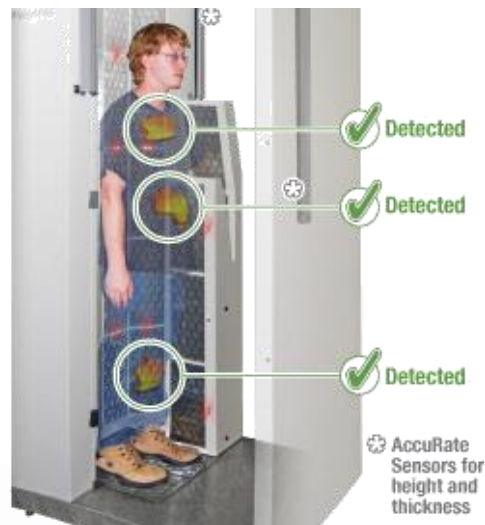
External contamination

Contamination meter

Measures external contamination of body, clothing or on the surface of objects and premises in Bq/cm², (devices coupled with probes α , β , γ)

Labile contamination

The search for labile contamination in Bq/cm² is carried out on dry surfaces using pads rubbed on a standardized surface. This smear is analyzed using a contamination meter with the corresponding probe.



Atmospheric beacon with alarm

Continuous monitoring (with threshold and alarm) or measurement of the concentration of radionuclides present in the atmosphere (suspended in the air) in Bq/m³. The measurement of atmospheric contamination (called "volumic activity") must be lower than the derived concentration limits in air defined according to the radionuclides present.

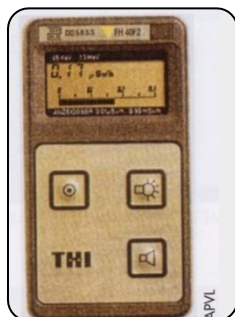
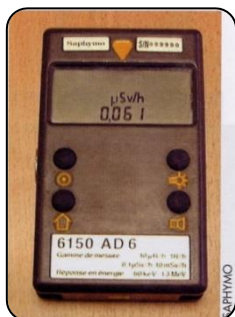
Principle: the air is sucked onto a filter which passes in front of a probe.



External exposure

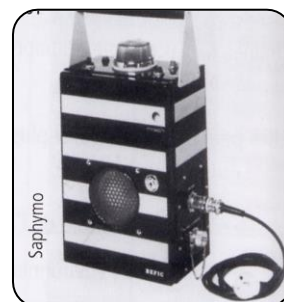
Flow meter

Measures an Equivalent Dose Rate in mSv/h (examples: 6150 AD6, FH40)



External Exposure Alarm Beacon

Monitors the Ambient Equivalent Dose Rate (in mSv/h)



8. Means of control and monitoring – 2/2

The internal exhibition

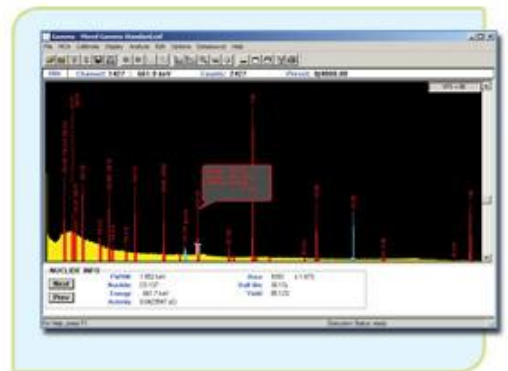
Anthropogammametry

Technique for physically measuring the radioactivity of the human body (natural or artificial radioactivity)



Radiotoxicology

Nuclear toxicology or radiotoxicology is an indirect method of measuring artificial radionuclides ingested, inhaled or having entered the human body through injury. For humans, this involves stool and urine analyses.



9. Humidity

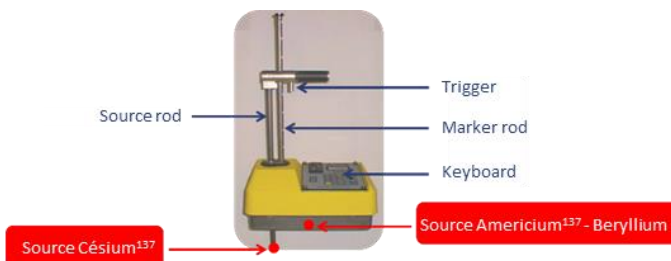
Equipment

Density and humidity measuring device containing radioactive sources

Used for quality controls in earthworks



Transport container. Class 7 dangerous goods – type A package



Required for employee users

- Passive and operational dosimetry
- Medical aptitude Category B (periodicity 2 years)
- Training
- Authorization by the project delegate
- Prohibited for minors, temporary workers, fixed-term contracts and pregnant women.



Required for transportation

- Driving license
- ADR and fire extinguisher awareness training (periodicity 3 years)
- Vehicle equipped with Transportation of Dangerous Goods
- Transport documentation
- Cellphone
- Compliant package and radiation detector

Apply any more restrictive local regulations



Required for storage

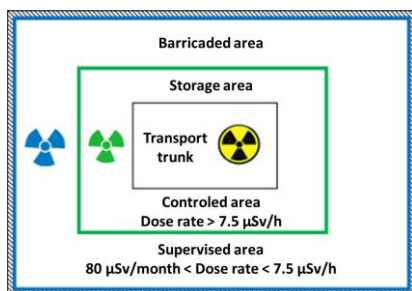
- Lock
- Anti-theft monitoring
- Regulatory display
- Extinguisher
- Zoning
- markup

Apply any more restrictive local regulations



Zoning

Storing a device



Operation area (dose rate greater than 2.5 µSv /h)

**CROSSING
FORBIDDEN
Secure area for
operation**

